

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (18 points)*Submit the postlab to the TA at the end of the lab.***Spectrometer Alignment and Calibration (5 points)****Question 3****3 points**

Fill in your zero angles and the grooves/mm written on the grating.

Data table 1: *Zero angles and diffraction grating size used in the spectrometer.*

Quantity	Value
Initial zero angle:	$\theta_{0,i} =$ _____
Final zero angle:	$\theta_{0,f} =$ _____
Grating grooves/mm:	_____ grooves/mm

Next, fill in the calibration data from the sodium D-lines:

Data table 2: *Calibration data from sodium D-lines.*

Spectral Line	Measured Angle	Difference Angle	Wavelength (nm)	Order m	Grating sep. d (nm)	$\bar{d}$ (nm)
Left, Line 1						
Right, Line 1						
Left, Line 2						
Right, Line 2						

Question 4**1 point**Show an example calculation of how you determined the grating spacing, d , using data from Data Table 2. Calculate an average value for d .

Question 5**1 point**

Calculate the uncertainty in grating separation, Δd , using the standard deviation of your two (or four) values for d .

The Balmer Series (13 points)**Question 6****2 points**

Fill in the table with your measurements of the Balmer lines.

Data table 3: *Balmer series measurements.*

Spectral Line	Order	Measured Angle	Difference Angle	Average Angle	m	Wavelength (nm)	Average Wavelength (nm)
Red, Right	1st						
Red, Left	1st						
Red, Right	2nd						
Red, Left	2nd						
Cyan, Right	1st						
Cyan, Left	1st						
Cyan, Right	2nd						
Cyan, Left	2nd						
Blue, Right	1st						
Blue, Left	1st						
Blue, Right	2nd						
Blue, Left	2nd						

Question 7**1 point**

Show the calculation for how you found the wavelength of the first-order blue line.

Question 8**1 point**

Show the calculation for how you found the wavelength of the second-order red line.

Question 9**2 points**

Calculate the uncertainty in wavelength $\Delta\lambda$ for the first-order cyan line using error propagation. Use the uncertainty in the grating separation that you determined earlier, and estimate a value for the uncertainty $\Delta\theta$, which you must convert to radians. Using the average angle θ when computing $\tan \theta$ in the expression below, compute $\Delta\lambda$:

$$\left(\frac{\Delta\lambda}{\lambda}\right)^2 = \left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta\theta}{\tan \theta}\right)^2. \quad (4)$$

Question 10**1 point**

Show the calculation for the theoretical Balmer wavelength for $n = 3$ using Table 1 and eq. (2).

Question 11**1 point**

Show the calculation for the theoretical Balmer wavelength for $n = 4$.

Question 12**1 point**

Show the calculation for the theoretical Balmer wavelength for $n = 5$.

Question 13**2 points**

Assume that the wavelength uncertainty $\Delta\lambda$ that you calculated for the cyan line also applies to the blue and red lines. Are the theoretical Balmer wavelengths consistent with your measured wavelengths within the uncertainties? Why or why not?

Question 14**2 points**

Ordinary hydrogen has a nucleus consisting of one proton. However, naturally occurring hydrogen contains a small amount of deuterium, hydrogen having a nucleus consisting of one proton and one neutron. If deuterium existed in sufficient quantity in your source to produce observable spectral lines, would you expect to resolve its lines from those of ordinary hydrogen with our equipment? Let the neutron mass be equal to the proton mass, and let the proton mass be about 1840 times greater than the electron mass. You are expected to show a calculation of some kind, not just an answer.